

Probabilistic Assessment and Forecasting of Landslide Occurrences using ML

Factor	Dataset Needed	Where to Request in Sri Lanka	Format	Resolution	Purpose
Slope Angle	DEM (LiDAR / high-resolution elevation)	Survey Department – Geoportal (ilas.survey.gov.lk/geoportal) or Map Portal (app.survey.gov.lk/mapportal). Higher-res DEM via Geoinformatics Branch: dsgeoinfo@survey.gov.lk	Raster (GeoTIFF)	<10m (requested); 30m fallback	Core susceptibility input; maps to NBRO risk bands (>35° Very High → 11–16° Low)
Soil Classification	Soil type maps (Red-Yellow Podzolic / Mountain Regosol)	Department of Agriculture (doa.gov.lk) and NBRO LRRMD geotechnical reports (info@nbro.gov.lk)	Vector (polygon)	District / mountain-block level	Identifies clay-trap zones (Badulla, Nuwara Eliya, Kandy) vs colluvium zones (Kegalle, Ratnapura)
Geology / Bedrock Type	Bedrock and weathering-depth maps	Geological Survey & Mines Bureau (GSMB) – request via office, gsmb.gov.lk	Vector (polygon)	District level	Structural weakness / deep weathering (e.g. fragmented granitic gneiss at Aranayaka)
Land Use / Land Cover	Land use classification, forest cover	Survey Department – topographic vector databases (survey.gov.lk)	Raster / Vector	10m or 1:50,000 base	Deforestation proxy – loss of root-anchoring correlates with failure
Hydrology / Drainage Network	Stream network, drainage lines	Survey Department – topographic data layers (survey.gov.lk)	Vector (line)	1:50,000 topo base	Proximity to drainage; saturation and channel-adjacent risk
Rainfall Intensity (real-time)	Rain gauge network + historical rainfall records	Department of Meteorology (meteo.gov.lk); NBRO automated rain gauges via LRRMD	Time-series (CSV/API)	Hourly / daily, per station	NBRO trigger thresholds: 75mm/24h Watch, 100mm/24h Alert, 150mm/24h or 75mm/1h Evacuation
Soil Saturation / Moisture	Soil moisture readings or piezometer data	NBRO LRRMD (instrumented sites) – info@nbro.gov.lk	Sensor time-series	Point-level, per site	Captures antecedent moisture and pore-water pressure buildup
Dangerous Slope Cutting / Building Footprint	Building locations, construction permits	Survey Department building layer (survey.gov.lk); local Divisional Secretariat / local authority permits	Vector (point/polygon)	Building-level	Flags artificial slope destabilization from excavation or construction
Population Exposure	Population counts by administrative division	Department of Census & Statistics (statistics.gov.lk)	Vector / tabular	Grama Niladhari division level	Weights risk score by number of people actually exposed
Infrastructure Accessibility	Road network data	Survey Department (survey.gov.lk); Road Development Authority (RDA)	Vector (line)	Road-segment level	Evacuation route viability and emergency-response prioritization
Historical Landslide Inventory / Hazard Zones	Landslide hazard zonation maps, past event records	NBRO LRRMD data portal (liris.nbro.lk/en); email info@nbro.gov.lk, subject: "Attn: Director, LRRMD - Landslide Dataset Request"	Vector polygon (hazard_type, zone_id, risk_score)	Zone level	Ground-truth labels for training and validating the ML model